

Opportunity Statement:

Legacy room acoustic calibrations (Audyssey, YPAO, Dirac) rely on reactive or delayed phase-based microphone sweeps that fail to perceive physical room architecture. *SONERAMIC* aims to disrupt the status quo with proprietary sonar hardware and AI software, creating a predictive digital twin of the room by scanning both architecture (LiDAR) and ultrasonic response simultaneously, banking on a two-sided network.

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Strategic Prioritization: Scaling the Voxel Data Flywheel

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Data Alignment:



The At-Cost Flywheel.

By deploying 10,000 units at cost (~\$15), we capture a proprietary library of 10,000+ architectural voxel meshes. This data moat makes our AI smarter and more predictive than legacy competitors.

Cultural Alignment:



Frictionless Luxury.

High-end enthusiasts pay a premium for perfection but desire simplicity. We replace complex legacy microphones with an intuitive 60-second scan that dominates the "prosumer" experience.

Success Metrics:



Volume over Margin.

Primary KPI is the Volume of Scans Received, establishing the "Ground Truth" needed for the global standard for acoustic valuation compounding flywheel momentum on both network sides.

Market Disruption:

The Reference Standard.

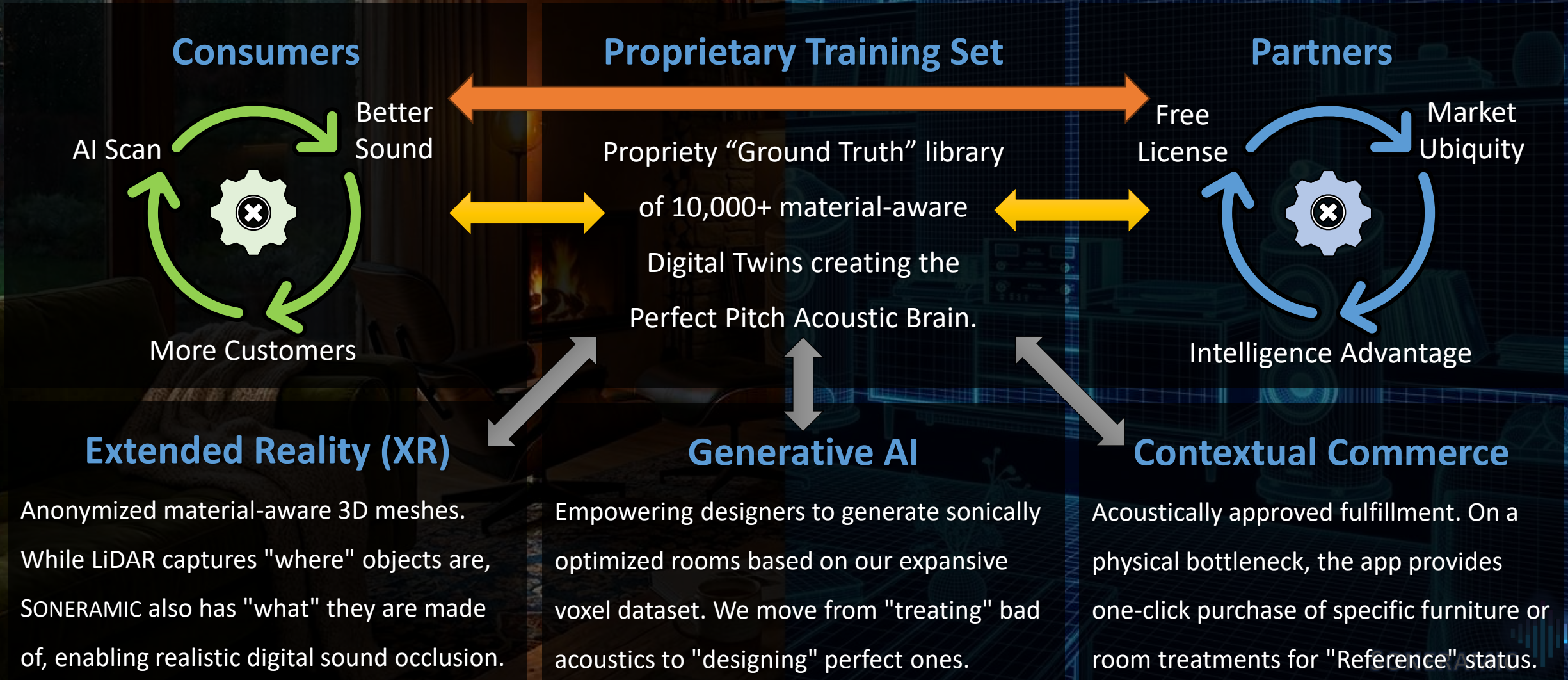
We commoditize legacy incumbents by offering our core algorithm as a free reference license to flagship AVR manufacturers (Denon, Sony, Harman). This ensures market ubiquity while securing a continuous stream of anonymized room telemetry for our "Acoustic Brain".



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Network Effects:

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Strategic Data Inventory:

Data Asset	Collection Method	Data Type	Strategic Utility	Data Asset
Architectural Voxel Mesh	Proprietary Sonar Hardware	3D Spatial Cloud	Acts as the Geometric Ground Truth; essential for physics-based modeling.	Architectural Voxel Mesh
Material Density Matrix	Ultrasonic Pulse-Echo Analysis	Numerical/Matrix	Material Ground Truth: Differentiates between soft (velvet/leather) and hard (glass/wood) surfaces.	Material Density Coefficients
Sonic Room Signature	Integrated Audio-Feedback Loop	Acoustic Waveform	Records how the specific room architecture reacts to sound at different frequencies.	Sonic Room Signature
Sonemeter Score (SABS)	Algorithmic Calculation	Normalized Benchmark	Creates a universal score for acoustic quality used for user benchmarking and social sharing.	Sonemeter Score (SABS)
AVR Hardware Metadata	OEM IoT Integration (Denon/Marantz)	System Telemetry	Ensures the algorithm is perfectly tuned to the specific power capabilities of the user's amplifier.	AVR Hardware Metadata

Current State Swim Lane:

Lane / Actor	Step 1 - Preparation	Step 2 - Execution	Step 3 - Analysis	Step 4 - Outcome
Customer	Manually setups microphone on a tripod in 8+ specific locations.	Initiate "Pink Noise" sweeps; remain silent for 20–45 minutes.	Review cloud-calculated graphs on a PC/Laptop.	Save and upload filters to the AVR.
Legacy Software		Capture delayed sound pressure levels from the microphone.	Calculate generic 1D frequency response curves (EQ).	Export static filters to hardware.
Hardware		Emits high-volume noise bursts through speakers.		Apply "blind" filters that ignore furniture and materials.

Transformed Swim Lane:

Lane / Actor	Step 1 - Initiation	Step 2 – Voxel Mapping	Step 3 - Analysis	Step 4 - Outcome
Customer	Open SONERAMIC App; hold smartphone for a 60-second "room sweep."		Review the Digital Twin mesh on phone.	Experience immediate "Reference" sound; check Sonemeter Score .
Ultrasonic Sensor	Activate proprietary ultrasonic pulse-echo sequence.	Simultaneously capture room geometry AND material density.		
Edge AI Engine		Construct a high-fidelity Voxel Mesh in real-time	Apply "Material-Aware" physics models (e.g., Glass vs. Wood).	Push predictive calibration profile to hardware.
AVR / Speaker				Performance is optimized based on the room's Ground Truth

Experimental Design: Validating the Voxel Advantage

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Objective: To prove that predictive voxel-mapping delivers a statistically significant improvement in acoustic accuracy compared to reactive legacy microphone sweeps.

Test Population: 250 randomized High-Net-Worth (HNW) households currently equipped with flagship audio systems (e.g., Bowers & Wilkins 800D Series or McIntosh).

Test Period: A 60-day longitudinal study monitoring long-term user satisfaction and acoustic performance delta.

Randomization: An A/B Crossover Design where Group 1 utilizes Legacy Dirac/Audyssey for the first 30 days before switching to **SONERAMIC**, while Group 2 utilizes the **SONERAMIC** Voxel Twin first, followed by Legacy.

Primary Hypothesis: Material-aware voxel mapping will achieve a 20% increase in the **Sonemeter Acoustic Benchmark Score** (SABS) compared to traditional 1D frequency-response correction.

Scaling Voxel Ecosystem:

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B2B Strategic Moat: "SONERAMIC Ready"

- Disrupt legacy incumbent licensing by offering the core algorithm as a Free License to flagship AVR manufacturers (Denon, Marantz, Sony).
- The Catch: Manufacturers must contribute anonymized room telemetry back to the **SONERAMIC** flywheel, cementing our position as the "pinnacle" of acoustic data.

Certification & Badging: The "Acoustic Zestimate"

- Monetize the **SONEMETER Score** as a standardized certification badge for luxury real estate and interior design.
- High-end home listings leverage the "**SONERAMIC GOLD CERTIFIED**" badge to prove room-perfect acoustics, adding tangible value to furniture and wall covering manufacturers.

Big Tech Data Licensing: Spatial Computing Infrastructure

- License anonymized material-aware 3D room meshes to Apple (Vision Pro) and Meta (Quest).
- While LiDAR captures "where" objects are, **SONERAMIC** captures "what" they are made of. This data is critical for realistic digital object occlusion and physics-based sound in AR/VR.

Lead-Gen & Affiliate: The "Voxel Marketplace"

- Direct affiliate integration with luxury partners (Eames, Roche Bobois, Knoll, ...).
- When the scan identifies "Reflective Hard Surfaces," the app offers one-click purchase for a specific acoustic panel or rug, earning 10–15% commission.